



WEEK 1 • DAY 4

DynareJulia

Implementing QPMs in a modern, open-source platform

01

From Dynare/Matlab to DynareJulia

Why the rewrite — and what changes

02

Nonlinearities & Occasionally Binding Constraints

ELB, MCPs, and dark-corner avoidance

03

Endogenous Credibility

Building credibility-aware models for FPAS Mark II



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What You'll Learn Today

Move yesterday's calibrated QPM into DynareJulia and solve the harder problems



DynareJulia in practice

Project structure, syntax, and how a .mod-style model becomes Julia code.



Nonlinearities

When linearisation fails — kinks, asymmetries, and credibility regimes.



Occasionally binding constraints

Solving the ELB and MCP-style problems that linear models cannot handle.



Endogenous credibility

Letting the model decide how much agents trust the central bank in each scenario.



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Why DynareJulia?

A modern rewrite — keeping what worked, fixing what didn't



Open & Free

No Matlab licence required.
Removes a real barrier for
emerging-market central banks.



Performant

Julia delivers near-C performance
— large nonlinear models become
tractable on a laptop.



Transparent Source

Solver internals are inspectable
Julia code, not black-box MEX files.



Reproducible Workflows

Package management and
versioning make every result
rerunnable years later.



Production-Ready

Designed to plug into reporting
pipelines, scheduling, and modern
dev tools.



Composable

Easy interop with Python, R, and
the wider Julia scientific stack.



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Dynare/Matlab vs DynareJulia

Same modeling philosophy — different platform, different ergonomics

	Dynare / Matlab	DynareJulia
Licensing	Matlab licence required	Free, open source
Performance (nonlinear)	Slow on large MCPs	Near-C speed via Julia
Solver transparency	MEX black box	Pure Julia, inspectable
Reproducibility	Manual version tracking	Project.toml / Manifest.toml
Pipeline integration	Limited	Native Python/R interop
Maturity & ecosystem	Decades of use, large library	Newer, modern scientific stack

Both tools are valid choices today. DynareJulia is built where institutional needs are heading: open, fast, reproducible, and integrated.



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A First Look: QPM in DynareJulia

What yesterday's QPM looks like in code — clean, declarative, recognisable

```
qpm.mod — DynareJulia

var y_gap, pi, i, e_gap;
varexo eps_y, eps_pi, eps_i;

parameters rho, beta, gamma, alpha,
            lambda, phi, theta_pi, theta_y;

rho      = 0.7;  beta      = 0.5;
gamma    = 0.2;  alpha    = 0.6;
lambda   = 0.3;  phi       = 0.8;
theta_pi = 1.5;  theta_y   = 0.5;

model;
  // IS curve
  y_gap = rho*y_gap(-1) - beta*(i - pi(+1)) + eps_y;
  // Phillips curve
  pi     = alpha*pi(+1) + (1-alpha)*pi(-1) + lambda*y_gap + eps_pi;
  // Policy rule
  i = phi*i(-1) + (1-phi)*(theta_pi*pi + theta_y*y_gap) + eps_i;
end;

stoch_simul(order=1, irf=20);
```

WHAT TO NOTICE

Familiar .mod syntax

If you know Dynare, you already know DynareJulia.

Equations look like the math

Output gap, Phillips, policy rule — each line maps to yesterday's slides.

Calibration in one block

Parameters set once, then the model runs — easy to swap for scenarios.

Same .mod files run in both

Migration is incremental, not a rewrite.



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Endogenous Credibility

Letting the model decide how much agents trust the central bank



THE BIG IDEA

Agents form expectations using a weighted mix of the central bank's target and observed past inflation. The weight (credibility) evolves with the bank's track record — so credibility itself becomes a state variable.

HIGH CREDIBILITY

Expectations Anchored

- Agents put high weight on the target
- Inflation deviations are read as transitory
- Policy can act gradually without losing control
- Shocks are absorbed by the system

LOW CREDIBILITY

Expectations Drifting

- Agents weight observed past inflation more
- Deviations become persistent and self-reinforcing
- Policy must act forcefully to regain control
- The cost of late action rises sharply



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Today's Hands-On Workflow

Four steps from yesterday's calibrated QPM to a scenario-ready DynareJulia model

01

Port

Translate yesterday's QPM equations into a DynareJulia .mod file. Same structure, different runtime.

02

Solve

Run linear simulation; verify steady state and impulse responses against the Matlab baseline.

03

Add nonlinearities

Introduce ELB constraint and endogenous credibility; resolve as an MCP.

04

Run scenarios

Generate Case A / Case B paths with credibility regimes — the inputs for tomorrow's report.

By end of day: a scenario-ready model. By end of week: a full FPAS Mark II run.



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Discussion & Wrap-up

Questions to take into Friday's capstone

KEY TAKEAWAYS FROM TODAY

- DynareJulia keeps the .mod modeling philosophy — but adds open licensing, speed, and reproducibility.
- Linear approximations break at the ELB and under regime shifts; MCP-based solvers handle the kinks.
- Endogenous credibility turns trust into a state variable — central to risk-management policy analysis.
- Today's port-and-extend workflow gives you a scenario-ready model for the Friday capstone.



DISCUSSION QUESTIONS

1. What would adopting an open-source modeling stack change in your institution's workflow?
2. Which historical episode in your country would benefit most from a nonlinear / ELB-aware model?
3. How would you measure credibility in your economy — what data would you reach for?
4. What governance steps are needed before a new model platform can produce decision-grade outputs?